

E7 Introduction to Computer Programming for Scientists and Engineers

Discussion week 14

Randomness and Monte Carlo simulation

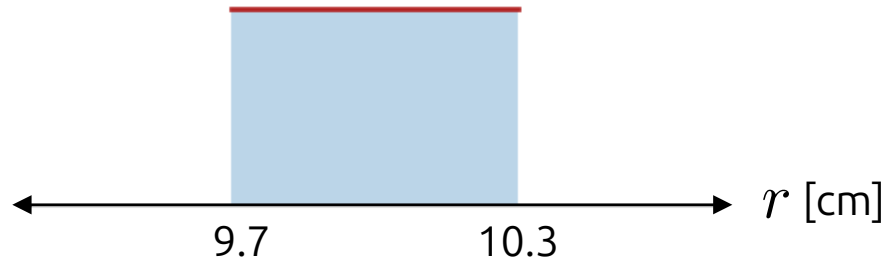
Systems we've encountered

System	Inputs	Solve	To obtain
spherical mixer	h	evaluate a function	total volume
heated plate	measured temperatures	system of linear equations	unmeasured temperatures
simple pendulum	$\ell, m, y(0), y'(0)$	IVP	energy after 20 swings
three tanks	$a, h_i(0), s(t), d_i(t)$	IVP	$\max_{t \in [0,1]} f_{12}(t)$

Modeling uncertainty with probability theory

- Random variable: A variable whose value is ***sampled*** from a ***distribution***.

Example: The radius r of the spherical mixer is *uniformly distributed* between 9.7 cm and 10.3 cm.

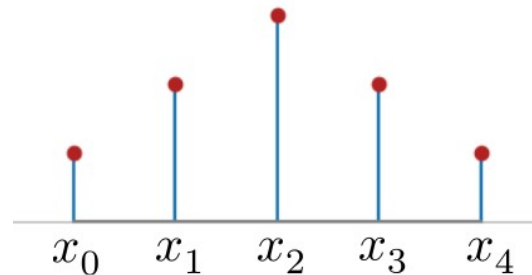


- A random variable has a **sample space** and a **distribution**.
- The **sample space** of a random variable is the set of all possible values.
- The **distribution** of a random variable is a function $p: X \rightarrow \mathbb{R}$ that encodes the sampling probabilities.
- The distribution has these properties:
 - Positivity: $p(x) \geq 0 \quad \forall x \in X$
 - Unit integral (or sum):

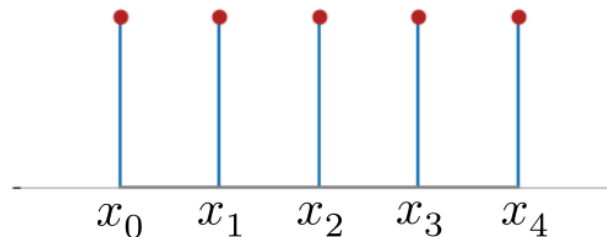
$$\int_X p(x) dx = 1 \quad \text{or} \quad \sum_{x \in X} p(x) = 1 \quad \text{if } X \text{ is discrete}$$

Some important types of random variables

- **Discrete r.v. :** X is a finite set.

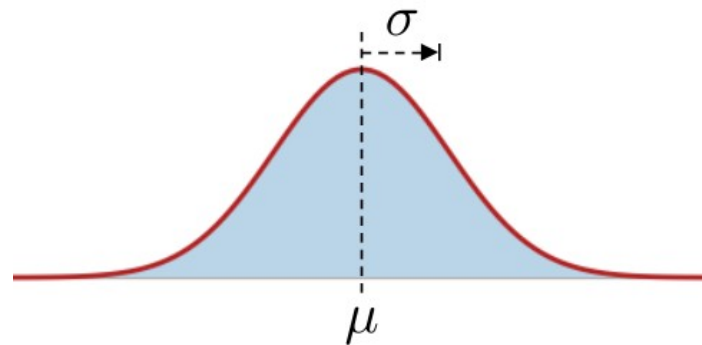


- **Uniform r.v. :** $p(x)$ is constant over X .



- **Gaussian (a.k.a. normal) r.v. :**

- $X = \mathbb{R}$.
- $p(x)$ is “bell-shaped”.

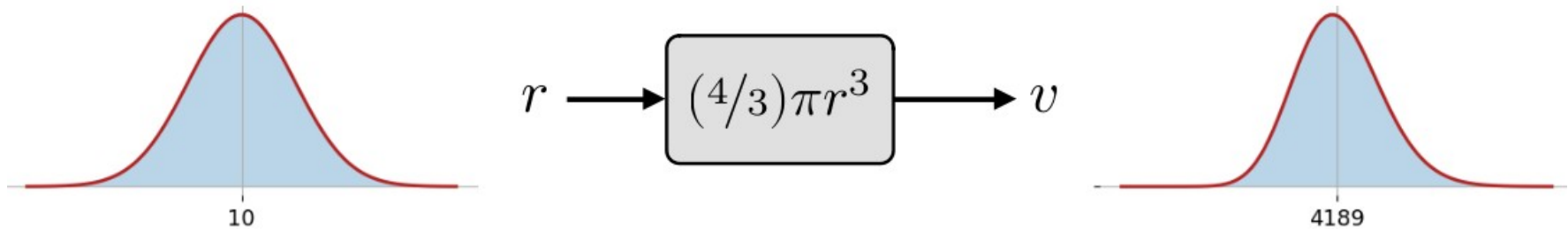


Functions of random variables

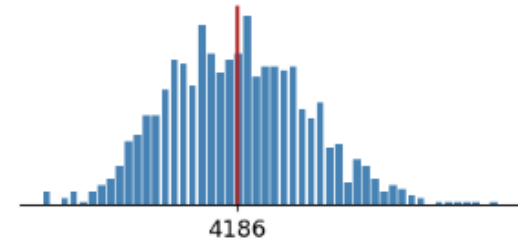
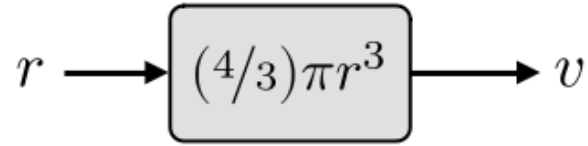
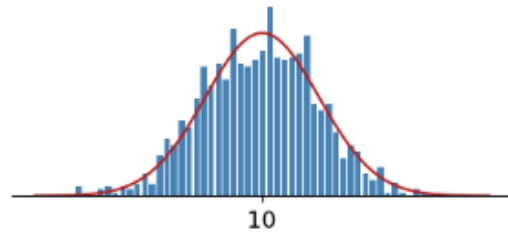
- The radius r of the spherical mixer is modeled as a uniform distribution between 97 mm and 103 mm.
- The total volume v of the sphere relates to r through

$$v = \left(\frac{4}{3}\right)\pi r^3$$

- What is the distribution of v ?



Monte Carlo simulation



In general

samples of
input values



samples of
output values

Sampling distributions with SciPy



- Create a random variable with a given distribution:

- Generic discrete `X = scipy.stats.rv_discrete(values=(X,p))`
- Continuous uniform `X = scipy.stats.uniform(loc=a,scale=b-a)`
- Gaussian `X = scipy.stats.norm(loc=mu,scale=sigma)`

- Sample the random variable:

```
X.rvs(size=<int>)
```

```
X.rvs(size=<int>,random_state=<int>)
```

... to ensure repeatability.